Week 1: Algebra Foundations

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September 5, 2025

Week 1 Quiz

1.1: Order of operations; integers & rational arithmetic; combining like terms

1.2: One & two step linear equations; translating verbal to algebra

1.3: Multi-step equations; distributing; like-term consolidation

Quick Reference: Equations, Definitions, & Formulas

Order of Operations (PEMDAS)

1. Parentheses

2. Exponents

3. Multiplication and Division (together, from left to right)

4. Addition and Subtraction (together, from left to right)

Properties of Real Numbers

- Commutative Property: $a + b = b + a; \quad ab = ba$
- Associative Property: $(a + b) + c = a + (b + c); \quad (ab)c = a(bc)$
- Distributive Property: $a(b + c) = ab + ac$
- Identity Property: $a + 0 = a; \quad a \cdot 1 = a$
- Inverse Property: $a + (-a) = 0; \quad a \cdot \frac{1}{a} = 1$ (for $a \neq 0$)

Combining Like Terms

Like terms have the same variable(s) raised to the same power(s).

To combine like terms, add or subtract their coefficients.

Week 1 Quiz: Algebra Foundations

Part A (1.1): Order of Operations (PEMDAS)

Instructions: Simplify the following expressions:

1. $9 + 3 \times 2$
2. $(6 + 8) \div 2$
3. $5^2 - (3 \times 2)$
4. $(18 \div 3) + (4 \times 2)$

Part B (1.2): Properties of Real Numbers

Instructions: Identify the property used in each equation:

1. $a + b = b + a$
2. $(2 + 5) + 7 = 2 + (5 + 7)$
3. $6 \times 1 = 6$
4. $3 \times (4 + 2) = 3 \times 4 + 3 \times 2$
5. $x + (-x) = 0$

Part C (1.3): Combining Like Terms

Instructions: Simplify the following expressions by combining like terms:

1. $4x + 7x$
2. $12a - 3a + 5$
3. $5n + 2m - n + 3m$
4. $7x - 2x + 6 - 3$

Part D: Word Problems

Instructions: Solve the following word problems using the concepts learned:

1. A bookshelf's width is $3x - 2$ inches and the length is $5x + 4$ inches.
   Write a simplified expression for the perimeter in inches.

2. A car rental company charges 35 per day plus a one-time 25 insurance fee.
   Write a simplified expression for the total cost of renting a car for x days.

Part E: Long Division Practice

Instructions: Perform the long division and show all work for the following problems:

1. Divide 648 by 3.

Bonus Challenge (Optional):

Instructions: Simplify the following expression:

1. $3(2x + 5) - 4x + 6$.

Solution Guide

Part A (1.1): Order of Operations (PEMDAS)

1. $9 + 3 \times 2$

Solution:
Multiply first (M before A in PEMDAS):
$9 + 3 \times 2 = 9 + 6 = 15$

$\boxed{\text{Answer: 15}}$

2. $(6 + 8) \div 2$

Solution:
Parentheses first, then divide:
$(6 + 8) \div 2 = 14 \div 2 = 7$

$\boxed{\text{Answer: 7}}$

3. $5^2 - (3 \times 2)$

Solution:
Parentheses first, then exponents, then subtract:
$5^2 - (3 \times 2) = 5^2 - 6 = 25 - 6 = 19$

$\boxed{\text{Answer: 19}}$

4. $(18 \div 3) + (4 \times 2)$

Solution:
Evaluate each parenthesis, then add:
$(18 \div 3) + (4 \times 2) = 6 + 8 = 14$

$\boxed{\text{Answer: 14}}$

Part B (1.2): Properties of Real Numbers

1. $a + b = b + a$

$\boxed{\text{Answer: Commutative Property of Addition}}$

2. $(2 + 5) + 7 = 2 + (5 + 7)$

$\boxed{\text{Answer: Associative Property of Addition}}$

3. $6 \times 1 = 6$

$\boxed{\text{Answer: Identity Property of Multiplication}}$

4. $3 \times (4 + 2) = 3 \times 4 + 3 \times 2$

$\boxed{\text{Answer: Distributive Property}}$

5. $x + (-x) = 0$

$\boxed{\text{Answer: Inverse Property of Addition}}$

Part C (1.3): Combining Like Terms

1. $4x + 7x$

Solution:
Both terms have the same variable x, so add the coefficients:
$4x + 7x = (4 + 7)x = 11x$

$\boxed{\text{Answer: } 11x}$

2. $12a - 3a + 5$

Solution:
Combine the a terms (the 5 is a constant and stays separate):
$12a - 3a + 5 = (12 - 3)a + 5 = 9a + 5$

$\boxed{\text{Answer: } 9a + 5}$

3. $5n + 2m - n + 3m$

Solution:
Combine the n terms and the m terms separately:
$5n + 2m - n + 3m = (5n - n) + (2m + 3m) = 4n + 5m$

$\boxed{\text{Answer: } 4n + 5m}$

4. $7x - 2x + 6 - 3$

Solution:
Combine the x terms and the constant terms separately:
$7x - 2x + 6 - 3 = (7 - 2)x + (6 - 3) = 5x + 3$

$\boxed{\text{Answer: } 5x + 3}$

Part D: Word Problems

1. A bookshelf's width is $3x - 2$ inches and the length is $5x + 4$ inches. Write a simplified expression for the perimeter in inches.

Solution:
Perimeter of a rectangle: $P = 2(\text{width}) + 2(\text{length})$
$P = 2(3x - 2) + 2(5x + 4)$
$P = 6x - 4 + 10x + 8$
$P = (6x + 10x) + (-4 + 8)$
$P = 16x + 4$

$\boxed{\text{Answer: } (16x + 4) \text{ inches}}$

2. A car rental company charges \$35 per day plus a one-time \$25 insurance fee. Write a simplified expression for the total cost of renting a car for x days.

Solution:
Total cost = (cost per day $\times$ number of days) + insurance fee
Total cost $= 35x + 25$

$\boxed{\text{Answer: } (35x + 25) \text{ dollars}}$

Part E: Long Division Practice

1. Divide 648 by 3.

Solution:

```
        216
      ------
    3 | 648
        6       (3 × 2 = 6)
        --
        04
         3      (3 × 1 = 3)
        --
         18
         18     (3 × 6 = 18)
         --
          0
```

Step by step:
- $6 \div 3 = 2$ (write 2 above)
- $4 \div 3 = 1$ remainder 1 (write 1 above, bring down 8 to make 18)
- $18 \div 3 = 6$ (write 6 above)
- Remainder $= 0$

$\boxed{\text{Answer: 216}}$

Bonus Challenge (Optional)

1. $3(2x+5)-4x+6$

Solution:
First, use the distributive property:
$3(2x+5)-4x+6$
$=6x+15-4x+6$

Now combine like terms:
$=(6x-4x)+(15+6)$
$=2x+21$

$\boxed{\text{Answer: } 2x+21}$